

Buy now above the fold

Rendering the Buy Now call-to-action above the fold on the product detail page, relative to the current below-fold placement, increases conversion_rate among product-page visitors on the demo storefront over the exposure window, with the expected direction being a higher per-user probability of placing at least one order. The treatment is hypothesized to act by reducing the interaction distance between product consideration and purchase initiation; the magnitude of the effect is pre-registered in this brief at the feasibility-bounded relative MDE below and is not asserted as a point prediction.

AT A GLANCE

Primary metric is conversion_rate; decision is binary against a positive lower bound on conversion lift AND no guardrail breach.

Pre-registered before any analysis. Decision is binary against conversion_rate: a 7.0% relative relative lift over a 10.0% baseline requires 29,718 per arm at the standard α / power settings. Guardrails: revenue_per_user, page_load_time, cart_abandonment_rate.

SAMPLE REQUIRED (PER ARM)

29,718

to detect a 7.0% relative relative effect at $\alpha = 0.05$, power = 0.80

BASELINE

10.0%

DURATION

**~7.7 days at
7,735/day
accrual**

TOTAL SAMPLE

59,436

FEASIBILITY

VIABLE

MECHANISM

Why we think this works

An above-fold Buy Now call-to-action reduces the steps between product consideration and purchase initiation; the treatment is expected to lift the per-user probability of placing at least one order.

Conversion baseline on the demo storefront sits near 10% over the trailing 28 days.

METRICS

conversion_rate is the verdict-driving metric.

Primary is conversion_rate; secondaries inform interpretation but do not influence the verdict. Guardrails block ship if their CI crosses the pre-registered non-inferiority margin in the adverse direction.

PRIMARY

conversion_rate

Proportion · baseline 10.0% · MDE 7.0% relative · direction higher is better

GUARDRAIL

revenue_per_user

Direction higher is better. Decision blocks if 90% CI lies past -1.0% relative.

GUARDRAIL

page_load_time

Direction lower is better. Decision blocks if 90% CI lies past 5.0% relative.

GUARDRAIL

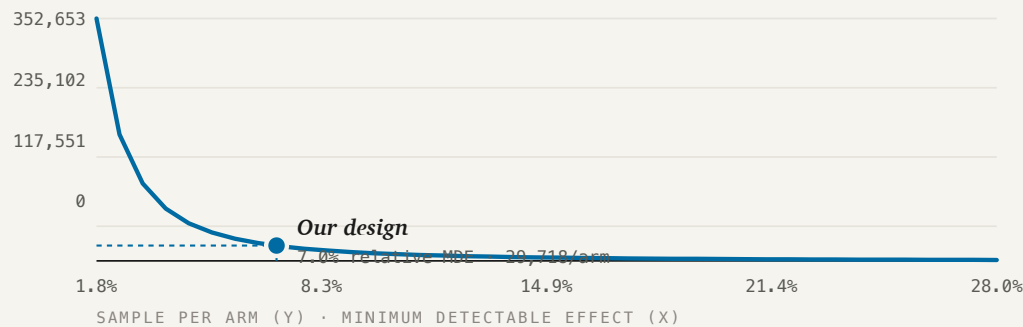
cart_abandonment_rate

Direction lower is better. Decision blocks if 90% CI lies past 5.0% relative.

POWER

Detecting smaller effects requires much more sample.

To resolve a 7.0% relative relative effect against a 10.0% baseline, the design requires 29,718 per arm. Tighter detection thresholds climb the curve to the left.



Smaller effects require disproportionately more sample. Our design point is feasible inside the available `user_id` surface.

ASSIGNMENT + COHORTS

First-exposure attribution at the `user_id` level.

Each user is randomly assigned at first exposure to control or treatment in a 50 / 50 split. Events before `first_exposure_at` are not attributed to the experiment.

UNIT	user_id One row per (<code>user_id</code> , <code>experiment_id</code>) in the assignments table; <code>first_exposure_at</code> gates which events are attributed.
ARMS	50 / 50 control · treatment
COHORT	Users on the demo storefront whose first exposure to the experiment is their first <code>page_event</code> row with <code>event_name = 'product_view'</code> during the exposure window; assignment fires at that event (<code>assignment.first_exposure_at = MIN(page_event.event_at WHERE event_name = 'product_view')</code> per <code>user_id</code>), so "all assigned users" is by construction the set of product-page visitors during the window, segment = general.

DECISION RULES

Pre-registered. The analyze verb walks the same tree against the same metrics.

Ship the treatment if and only if the two-sided 95% confidence interval on the relative lift in `conversion_rate` excludes zero on the positive side (lower bound strictly greater than zero) AND no guardrail breaches its pre-registered non-

inferiority margin. If the primary interval crosses zero, or if any guardrail's confidence interval crosses its non-inferiority margin in the adverse direction, the verdict is do-not-ship regardless of point estimate. The rule is intentionally tight: a treatment whose true effect is below the pre-registered MDE is expected to fail it.

SHIP	Primary 95% CI excludes zero on benefit side · no guardrail breach · novelty late-ratio ≥ 0.7
LIFT WITH CAVEAT	Real lift but below MDE/2 — surface the caveat in the readout but still consider rolling out
NO-SHIP · GUARDRAIL	Any guardrail's 90% CI lies past its harm threshold
NO-LIFT	Well-powered, primary CI straddles zero with half-width $> 2 \times \text{MDE}$
INCONCLUSIVE	Underpowered ($n < \text{required}$) AND primary CI straddles zero
INVALID	SRM $\chi^2 p < 0.0005$ and no override resolved — investigate assignment

INTEGRITY LOCK · THREE-PART SEAL

brief.sealed.yaml binds intent + metric definitions + expected shape.

Tampering with any of the three components after seal will fail

agentxp.schemas.brief_seal.verify_or_raise; agentxp analyze refuses to open a brief whose seal does not verify (R11). This is the architectural wall, not a guideline.

```
DESIGN_CHAIN_HASH
37770e1e22d1d584de8602b37156d00854a1c79702a1cd12e0829393849b1a13
METRIC_SNAPSHOT          4 metrics hashed
CONVERSION_RATE
f775e246e2992c91fa7a04f5d22bb93d2cb9ed5078144479b7792e6cdf9a5fe1
REVENUE_PER_USER
c4742f81560b2836b851f2c17bdd1bd3cb102fdf14a69a04cd9170b3cf9d69ae
PAGE_LOAD_TIME
f1914747b5fdf7aac000b3991e59262b5871c3fe85bfc18a82e8bfc7084563e4
CART_ABANDONMENT_RATE
6498a159e387655f28f9949108bee2f9954e7078afa745f6f0430c2c5eb36db5
EXPECTED_SHAPE           user_id · 50 / 50
```

SEALED_AT	2026-06-05T18:43:44.845392Z
SEALED_BY	shane@aieval.ai

BRIEF.SEALED.YAML SEAL 37770E1E22D1... · 2026-06-05T18:43:44.845392Z · R1 R11